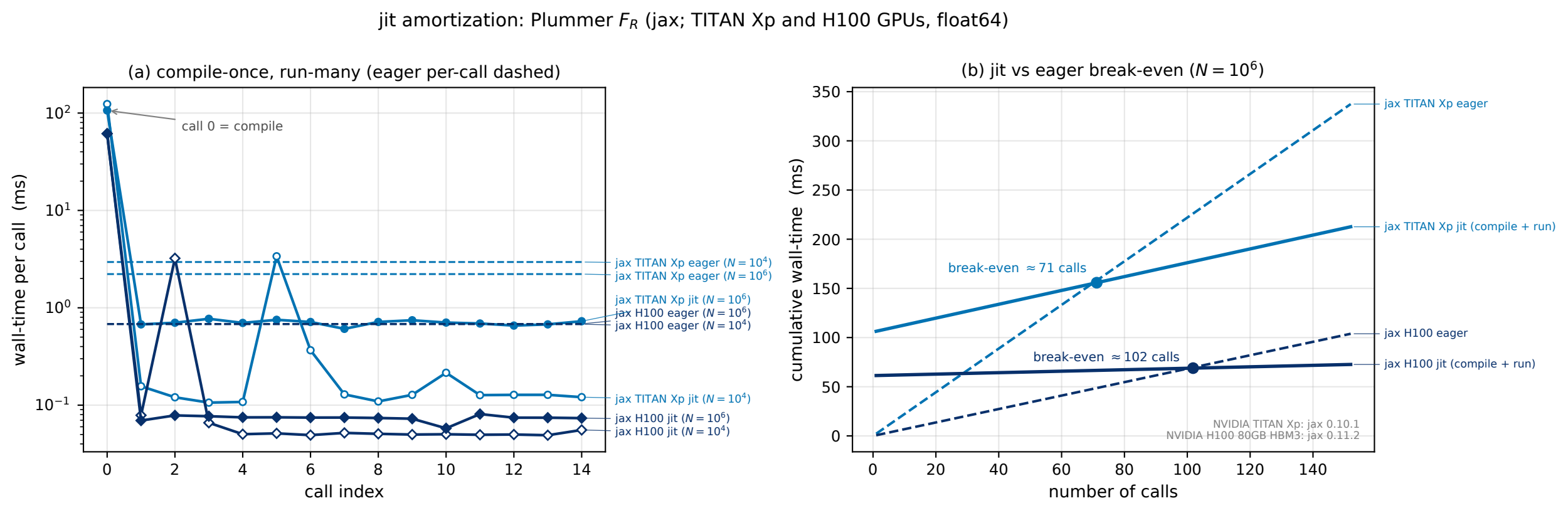
jit amortization: Plummer F_R (jax; TITAN Xp and H100 GPUs, float64)
(a) compile-once, run-many (eager per-call dashed)
wall-time per call (ms)
call index
call 0 = compile
jax TITAN Xp eager ($N = 10^4$)
jax TITAN Xp eager ($N = 10^6$)
jax TITAN Xp jit ($N = 10^6$)
jax H100 eager ($N = 10^6$)
jax H100 eager ($N = 10^4$)
jax TITAN Xp jit ($N = 10^4$)
jax H100 jit ($N = 10^6$)
jax H100 jit ($N = 10^4$)
(b) jit vs eager break-even ($N = 10^6$)
cumulative wall-time (ms)
number of calls
break-even ≈ 71 calls
break-even ≈ 102 calls
jax TITAN Xp eager
jax TITAN Xp jit (compile + run)
jax H100 eager
jax H100 jit (compile + run)
NVIDIA TITAN Xp: jax 0.10.1
NVIDIA H100 80GB HBM3: jax 0.11.2